04/25/2022

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M168_Notes_5

Section 5.1 Compound Interest

Simple interest – interest computed as a percentage of the principal

1% interest

$100 → $101 After a yr.

Simple interest earned: $I = Prt$

I=interest earned
P=principal
r=interest rate (decimal form)
t=time period (in years)

Future value: $F = P + Prt = P(1 + rt)$

Example
If a bank loans $525 to an individual for 3½ years at 7.25% simple interest, what will be the amount repaid on the loan?

$P = 525$
$r = 0.0725$
$t = 3.5$

$F = 525(1 + 0.0725 \times 3.5)$
$= 658.22$

Example
If a $1000 deposit grows in value to $1024 after 9 months, what is the simple interest rate that is earned?

$P = 1000$
$r =$ unknown
$t = 0.75$

$1000(1 + r \times 0.75) = 1024$

$1000 + r \times 0.75 \times 1000 = 1024$

$r = \frac{24}{0.75 \times 1000} = \frac{24}{750} = 3.2$

$r = 3.2\%$

Example

How much should be place in an account that pays simple interest of 4% so that the value of the account after 18 months is $3000?

$P =$ Unknown
$r = 0.04$
$t = 1.5$

$$P(1 + 0.04 \times 1.5) = 3000$$
$$\Rightarrow P = \frac{3000}{1 + 0.04 \times 1.5} = \boxed{2830.189}$$

Example

You have an account with $500 that pays 5% simple interest. How long until your account doubles in value?

$P = 500$
$r = 0.05$
$t =$ Unknown

$$P(1 + 0.05\,t) = 2P$$
$$\Rightarrow 0.05\,t = 1$$
$$\Rightarrow t = \frac{100}{5} = \boxed{20} \leftarrow \text{20 years.}$$

When interest is paid periodically and the interest earns interest, we have compound interest.

Example

You deposit $100 into an account that pays 10% annual interest that is compounded annually. How much is in the account after 4 years?

$$F_1 = 110$$
$$F_2 = 110(1 + 0.1) = 121$$
$$F_3 = 121(1 + 0.1) = 133.1$$
$$F_4 = 133.1(1 + 0.1) = 146.41$$

$m=1$

$$F = P\left(1+\frac{r}{m}\right)^{mt}$$

where m is the number of compounding periods per year and r is the annual interest rate as a decimal.

Example

You deposit \$300 into an account that pays 6% annual interest compounded monthly. How much is in the account after 3 months?

$P = 300$

$r = 6\%$

$m = 12$

$t = \frac{3}{12}$

$$F = 300\left(1+\frac{0.06}{12}\right)^{12\times\frac{3}{12}}$$

$$= 300(1+0.005)^3$$

$$= 304.52.$$

If interest is compounded	Then m is
Annually	1
Semiannually	2
Quarterly	4
Monthly	12
Weekly	52
Daily	365

ON THE CALCULATOR, TVM Solver:

```
N=
I%=0
PV=0
PMT=0
FV=1
P/Y=1
C/Y=1
PMT:END BEGIN
```

N is # of payments

I% is interest rate as percent.

PV is Present Value

PMT is Payments

FV is Final Value

P/Y and C/Y is # of payments or compounding period per year.

set PMT: END (make payments at the end of the cycle)

Example

You deposit \$100 into an account that pays 10% annual interest that is compounded annually. How much is in the account after 4 years?

$$F = 100\left(1 + \frac{0.1}{1}\right)^4 = 146.41$$

Example

You deposit \$300 into an account that pays 6% annual interest compounded monthly. How much is in the account after 3 months?

N = 3 PV = − 300 FV = ?

I = 6 PMT = 0 P/Y = 12

C/Y = 12

Example

You deposit \$1000 into an account that pays 10% annual interest. How much is in the account after 10 years if

(a) the account earns simple interest?

$F = 1000(1 + 0.1 \times 10) = 2000$

(b) the account earns interest compounded annually?

$N = 10$ $PMT = 0$

$I = 10$ $FV = ?$

$PV = -1000$ $PY = 1$, $CY = 1$

$\boxed{\$2593.74}$

(c) the account earns interest compounded semiannually?

$N = 20$

$PY = 2 = CY$

$\boxed{\$2653.30}$

(d) the account earns interest compounded quarterly?

$N = 40$

$PY = 4 = CY$

(e) the account earns interest compounded monthly?

(f) the account earns interest compounded weekly?

(g) the account earns interest compounded daily?

$N = 3650$

$PY = 365$, $CY = 365$

$\boxed{\$2717.97}$

Effective interest rate r_{eff} can be found with a FINANCE command

Eff(I%, m)

Example

You find a bank that pays 8.15% annual interest compounded quarterly and another that pays 8.1% annual interest compounded daily. Which is a better deal?

Eff(8.15, 4) = 8.402

Eff(8.1, 365) = 8.436 * Better deal

If interest is compounded continuously, then the future value of an investment of P dollars at an annual interest of r (expressed as a decimal) for t years is $\boxed{F = Pe^{rt}}$.

Example

You deposit \$1000 into an account that pays 10% annual interest. How much is in the account after 10 years if the interest is compounded continuously?

$1000 \cdot e^{(10 \times 0.1)} \approx 2718.28$

Suppliment calcusus heavy.

$$F = P\left(1 + \frac{r}{m}\right)^{mt}$$

$$\lim_{m \to \infty} \left(1 + \frac{r}{m}\right)^{mt} = e^{rt}$$

Example

You wish to have $60,000 available for your child's college expenses. You find an account that pays 7% annual interest compounded monthly.

How much needs to be deposited now to reach this goal if you want to have the money available in 18 years?

N = 18×12 = 216
I = 7
PMT = 0
FV = 60000
P/Y = 12
C/Y = 12

PV = 17081.66

How much interest was earned?

60,000 − 17,081.66 = 42,918.34

What if you wanted the money available in 8 years?

N = 12×8 = 96, Everything else remains same

PV = 34,328.35

How much interest was earned?

25,671.65

Example

You deposit $4000 in an account that earns 5% annual interest compounded weekly. How long does it take for the account to double in value?

N = ?
I = 5
PV = −4000
PMT = 0
FV = 8000

P/Y = 52
C/Y = 52

→ N = 721.22

⇒ $t = 721.22/52 = 13.87$ years.

Note: PV = −4000 is not a necessary information here.

Example

You deposited $200 in a savings account 10 years ago. It is now worth $350. You know the account paid annual interest that was compounded annually, but you don't remember the interest rate promised. What was the annual interest rate for the account?

N = 10
I = ?
PV = −200
PMT = 0

FV = 350
P/Y = 1
C/Y = 1

$\boxed{I = 5.76}$

Example

Your retirement account was worth $16,000 on January 1 and worth $15,000 two years later. Find the annual interest rate.

N = 2
I = ?
PV = −16,000
PMT = 0
FV = 15,000

P/Y = 1
C/Y = 1

I = −3.18 %

Example

To save for retirement, you deposit $2000 every year in an account that earns 6% annual interest compounded annually.

(a) How much will you have after 10 years? How much interest did you earn?

N = 10
I = 6
PV = 0
PMT = -2000

FV = ?
P/Y = 1
C/Y = 1

FV = 26,361.59

Interest = (26,361.59 − 10×2000)
= $\boxed{6,361.59}$

(b) How much will you have after 25 years? How much interest did you earn?

All same, N = 25 instead of 10.
FV = 109,729.02, Interest = (109,729.02 − 25×2000)
= 59,729.02

(c) How much will you have after 40 years? How much interest did you earn?

All same except N = 40.
FV = 309,523.93, Interest = (309,523.93 − 80,000)
$\boxed{= 229,523.93}$

An ***annuity*** is an account to which regular payments are made. An annuity that is ***certain and simple*** has the following properties:

1. The payments are made at fixed time intervals
2. The periodic payments are of equal size
3. The payments are made at the end of the interval
4. The interest is paid at the end of the interval

Many loans and savings plans are certain and simple annuities

Any account that is established to meet a future need is called a ***sinking fund***.

Example
You wish to have $60,000 available for your child's college expenses. You find an account that pays 7% annual interest compounded monthly. How much needs to be deposited every month to reach this goal if you want to have the money available in 18 years? Assume that the monthly payments are of equal size. How much interest does the account earn?

$N = 18 \times 12 = 216$
$I = 7$
$PV = 0$
$PMT = ?$
$FV = 60{,}000$

$P/Y = 12$
$C/Y = 12$

$\boxed{PMT = -139.30}$

Interest earned $= (60{,}000 - 139.3 \times 216) = \boxed{29{,}911.2}$.

Example
You start a savings account with the goal of saving $5000 in 4 years by making regular quarterly payments to an account that earns 8% annual interest compounded quarterly.

$N = 16$
$I = 8$
$PV = 0$
$PMT = ?$
$FV = 5000$
$P/Y = 4$
$C/Y = 4$

(a) How large are the quarterly payments?
268.25

(b) How much interest is earned in all?
$5000 - 268.25 \times 16 = 708.$

(c) How much interest is earned in the 3rd quarter of the 2nd year?

$\underbrace{1\ 2\ 3\ 4}_{\text{1st yr}}\ \underbrace{5\ 6\ 7\ 8}_{\text{2nd yr}}$ ← 3rd quarter

$N = 7$
$I = 8$
$PV = 0$
$PMT = -268.25$
$FV = ?$
$P/Y = 4$
$C/Y = 4$

$FV = 1994.25$, Interest $= 1994.25 \times \frac{0.08}{4}$
$= \boxed{39.89}$

Example
A cello costs $574 to buy or $40.58 per month on a 2 year rent-to-own plan. What is the interest rate charged?

Example
You owe $6000 on a credit card that charges 21% annual interest compounded monthly on the unpaid balance.

(a) If you make monthly payments of $120, how long will it take to pay it off?

$N = ?$
$I = 21$
$PV = 6000$
$PMT = -120$
$FV = 0$
$P/Y = 12$, C/Y

$N = 119.86$ months.

(b) How much interest is paid in all?

$I = 21$ $PMT = -120$ $P/Y = 12,\ C/Y$ $N = 119.86$ months.

(b) How much interest is paid in all?

Interest $= 120 \times 119.86 - 6000 \approx \$8383.2$

(c) How much [redacted] do you ~~owe~~ owe after 5 years of making payments?

$N = 60$ $PV = ?$ $FV = 0$ $PV = 4435.68$

$I = 21$ $PMT = -120$ $P/Y = 12$

Example

You want a pair of diamond earrings that cost \$800. You decide to stop buying daily lattes and instead save the \$5 every day in an account that pays 6% annual interest compounded daily. How long until you have the money for the earring?

$N = ?$ $PMT = -5$ $N \approx 158$ days

$I = 6$ $FV = 800$

$PV = 0$ $P/Y = 365$

Example

When you retire you would like to have \$30,000 per year. Assuming that you will need to have 20 payments starting right now, how much do you need to deposit into an account that pays 5% annual interest that is compounded annually?

$N = 20$ $P/Y = 1$

$I = 5$ $C/Y = 1$

$PV = ?$

$PMT = 30{,}000$ $PV = -373{,}866.31$

$FV = 0$

Example

You wish to purchase a house that costs \$130,000. You make a down payment of \$20,000 and finance the remainder for 30 years at 5.1% annual interest compounded monthly on the unpaid balance.

(a) How large are the monthly payments?

$N = 12 \times 30 = 360$ $P/Y = 12$ $PMT = \$597.24$

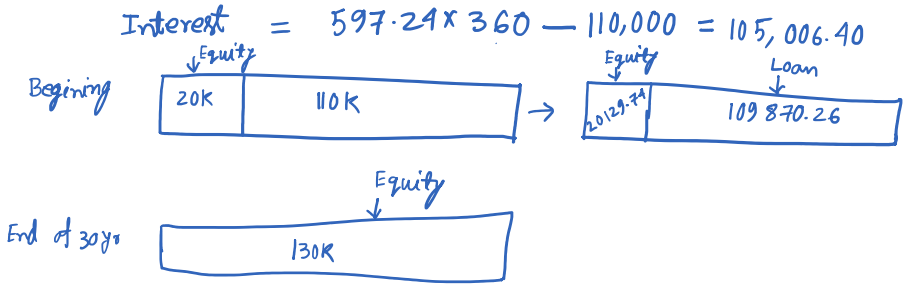
$I = 5.1$ $C/Y = 12,$

$PV = 110{,}000$

$PMT = ?$

(b) How much interest is paid in all?

Interest $= 597.24 \times 360 - 110{,}000 = 105{,}006.40$



(c) How much of the first payment is interest?

$597.24 > 110{,}000 \times \frac{0.051}{12} = \boxed{467.50}$

(d) How much do you still owe the bank after your first payment?

$110{,}000 - (597.24 - 467.5) = 109{,}870.26$

(e) What is your equity after your first payment?

$20{,}000 + (597.24 - 467.5) = 20{,}129.74$

(f) What is your equity after your second payment?

Interest $= 109{,}870.26 \times \frac{0.051}{12} = 466.95$

$597.24 - 466.95 = 130.29$

$109{,}870.26 - 130.29 = 109{,}739.97$

Equity $\rightarrow 130{,}000 - 109{,}739.97 = 20{,}260.03$

(g) What is your equity after your 12th payment?

N = 12
I = 5.1
PV = 110,000
PMT = -597.24
P/Y = 12

FV = -108,406.15

Equity: $130{,}000 - 108{,}406.15 \approx 21{,}593.85$

After 20 yrs
N = 240

$\rightarrow$ FV = 56052.36

Equity after 20 yrs = $130{,}000 - 56052.36$

Example

You borrow $3000 at 12% annual interest compounded quarterly for 2 years. Show in a table how much of each payment is interest, how much is paid towards the principal, how much of the loan is paid off and how much is still owed each quarter.

N = 8
I = 12
PV = 3000
PMT = -427.37
FV = 0
P/Y = 4

$3000 \times \frac{0.12}{4} = 90$

$2662.63 \times 0.03 = 79.88$

$2315.14 \times 0.03 = 69.45$

end of period	# PMT left	PMT ($)	interest ($)	towards principal	outstanding principal	equity
0	8				3000	0
1	7	427.37	90	337.37	2662.63	337.37
2	6	427.37	79.88	347.49	2315.14	684.86
3	5	427.37	69.45	357.92	1957.22	1006.78
4	4	427.37	58.72	368.65	1588.57	1375.43

3	5	427.37	69.45	357.92	1957.24	1006.78
4	4	427.37	58.72	368.65	1588.57	1375.43
5	3	427.37	47.66	379.71	1208.86	1755.14
6	2	427.37	36.27	391.10	817.76	2146.24
7	1	427.37	24.53	402.84	414.92	2549.08
8	0	427.37	12.45	414.92	0	3000 ←

Example

For the house purchased earlier, fill in the following lines of an amortization table.

end of period	# PMT left	PMT ($)	interest ($)	towards principal	outstanding principal	equity
0						
1						
2						
12						
60						
120						
180						
240						
300						